BVS1 · Winter Semester 2026/2027 · TH Köln · ICCT

Handout — Development Environment

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This handout collects commands that are useful after the initial Lab 0 setup. Use Lab 0 for the linear setup path; use this sheet for everyday operation, backup, and recovery.

Everyday VM Commands

Run these on the physical host:

```
multipass shell bvs1
multipass stop bvs1
multipass start bvs1
multipass info bvs1
```

Inside the VM, the course repository is:

```
cd ~/teaching-bvs1-code
```

Keep the VM Updated

Update the VM packages regularly, for example once a week. Inside the VM:

```
sudo apt update
sudo apt upgrade
```

If a restart is required, run `multipass restart bvs1` on the physical host. Afterwards, run `make check` in `~/teaching-bvs1-code`. Stay on Ubuntu 26.04 LTS: do not run `do-release-upgrade`. Avoid updating immediately before a checkpoint appointment.

Back Up Your Work

Local Git history is not a backup because the repository is stored inside the `bvs1` VM.

If you already use Git and GitHub, you may fork the public course repository, clone your fork inside the VM, and use the usual commit/push workflow to keep your work in your own remote repository. This is optional; the course does not require a GitHub account or a fork.

If you do not want to use GitHub, create a Git bundle and copy it out of the VM.

Inside the VM:

```
cd ~/teaching-bvs1-code
git bundle create ~/teaching-bvs1-backup.bundle --all
```

Return to the physical host:

```
exit
```

Copy the bundle out of the VM:

```
multipass transfer \
  bvs1:/home/ubuntu/teaching-bvs1-backup.bundle .
```

On Windows, Multipass may then print an error similar to:

```
[sftp] cannot set permissions for local file
.\teaching-bvs1-backup.bundle
```

This message concerns setting Unix-style file permissions on the Windows destination. If the bundle exists on the physical host and has a non-zero size, the transfer itself succeeded. In PowerShell you can check:

```
Get-Item .\teaching-bvs1-backup.bundle
```

If the file is shown and has a non-zero size, the transfer succeeded. Keep the bundle outside the VM, preferably in a location that is backed up regularly, such as your Sciebo folder or another backup location.

Troubleshooting

Stop at the first broken layer. Do not continue with later commands on the wrong machine.

Problem	Check
Multipass cannot read `bvs1-cloud-init.yaml`	Run `pwd` and `ls -l bvs1-cloud-init.yaml*` on the physical host. If the file ends in `.sec` (added by ILU), rename it to `bvs1-cloud-init.yaml`.
`multipass shell bvs1` says the instance does not exist	Create the VM first. Do not continue with Git or build commands on the physical host.
`cloud-init status --wait` reports an error	Run `cloud-init status --long` and `sudo tail -n 100 /var/log/cloud-init-output.log` inside the VM.
`make check` reports a missing tool	Verify `cloud-init status --wait` and `hostname`. Do not install arbitrary replacement packages.
`Hello BVS1` appears but no prompt returns	This is expected. Terminate QEMU with Ctrl-a, release, then x.

Recreate the VM During Initial Setup

If provisioning fails during Lab 0 before you have started course work, you may recreate the VM from the physical host:

```
multipass delete --purge bvs1
```

Then repeat the Lab 0 launch step.

You may also recreate the VM later in the semester if necessary. Before deleting it, back up the course repository stored inside the VM; `multipass delete --purge bvs1` permanently removes that local repository together with the VM.